

# Kenneth W. Bingham

AI Engineer | Full-Stack | Backend & Platform Engineer

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## SUMMARY

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Software engineer with 10+ years building reliable systems end to end. Full-stack from database to cloud deploy, with 4 years building and running Azure DevOps CI/CD pipelines. Focused on the reliability layer around large language models: deterministic pipelines, evaluation harnesses, structured-output validation, and cost-aware model routing. Comfortable owning a problem from architecture through production deployment, with a track record of shipping tested, reproducible systems. Seeking an AI engineering or backend/platform role where reliability and cost-efficiency matter.

## CORE SKILLS

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**Languages:** JavaScript / TypeScript, Python, SQL, C# / .NET, Java, GLSL

**Front-end:** Angular, React, WebGL2, HTML / CSS

**AI / LLM:** LLM integration, RAG, prompt engineering, evaluation harnesses, structured outputs, tool / function calling, model routing, prompt caching, token budgeting

**Backend & Infra:** Node.js, Express, REST, WebSockets, PostgreSQL, SQL Server, SQLite, Linux, nginx, Docker, PM2, zero-downtime deploys

**CI/CD & Cloud:** Azure DevOps pipelines (4 years), GitHub Actions, Azure

**Practices:** Event sourcing, content-addressed storage, deterministic replay, automated testing, observability

## SELECTED ENGINEERING PROJECTS

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### bfx-ingest - Deterministic LLM Context Tool (Open Source)

- Built a zero-dependency CLI that assembles deterministic, content-addressed context from a codebase, deduplicating by content hash to cut token usage and produce a reproducible root hash for pinned evals and prompt-cache hits.
- Emits model-specific formats (Markdown / XML / JSON) with token estimates; covered by automated tests with CI across four Node.js versions; MIT licensed.

### Zero-Downtime Deployment System

- Designed an atomic, content-addressed release pipeline: each deploy is hashed into a verifiable manifest, promoted by an atomic symlink swap, health-checked, and automatically rolled back on failure.
- Replaced in-place rsync deploys (which left the site broken on partial failure) with immutable releases and one-command rollback; integrated with GitHub Actions CI/CD across multiple domains.

### Deterministic Multiplayer Game Platform - KensGames

- Built real-time multiplayer netcode using event sourcing: game state is derived from a seed and an append-only command log, so every client replays identical events and converges on byte-identical state, eliminating an entire class of state-divergence bugs.
- Full-stack ownership: WebGL2 / Three.js rendering, Node.js / WebSocket backend, deployed and operated on a self-managed Linux VPS.

### ButterflyFx - Reproducible State Runtime (Open Source)

- Built a small event-sourcing runtime that wraps any reducer to provide content-addressed, hash-chained, tamper-evident state with deterministic replay, making reproducibility and auditability architectural properties; each property covered by tests.

## EXPERIENCE

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### Independent Software Engineer | 2020 - Present

- Designed and shipped end-to-end systems solo: a deterministic multiplayer platform, a reproducible CI/CD deployment pipeline, and open-source developer tools (see projects above).
- Concentrated on the reliability, reproducibility, and cost-efficiency of LLM-backed systems.

[FILL: Title] | [FILL: Company] | [FILL: start - end]

- [FILL: a concrete accomplishment in standard terms, with a number if possible]
- [FILL: a second accomplishment showing scope or ownership]

## CONCEPTUAL INITIATIVES

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Independent research in geometric and representation-based computation:

- **Dimensional programming / manifold-as-data:** deriving state and structure from low-dimensional latent representations and a shared seed, the same intuition as the manifold hypothesis, embeddings, and autoencoders. The design lens behind the deterministic, reproducible systems above (event-sourced netcode, content-addressed tooling).

## PRACTICAL APPLICATIONS

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What these systems make possible, each demonstrated in the projects above:

- Reproducible AI evaluations (exact context pinned by hash) and major LLM cost savings (byte-identical context maximizes prompt-cache hits).
- Provable AI provenance and tamper-evident records (hash-chained history where altering the past is detectable).
- Zero-divergence distributed and multiplayer state from a single shared seed; time-travel debugging and instant, self-healing deploys.

## INTERESTS

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Drummer, singer, and acrylic painter; independent researcher.

## EDUCATION

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[FILL: Degree or relevant coursework] | [FILL: School] | [FILL: year]